MAE 3127 Engineering Ethics Scenarios (Fall 2025)
Note: Each group will be assigned one case scenario

Scenario #1: You are a product quality control engineer at a company that produces airbags for many major automotive companies like Honda, Toyota, Ford etc. Your analysis of accident reports shows that there is a greater frequency of accidental airbag activation for cars that are in hot and humid environments. It is also noted that airbags that activate accidentally usually create dangerous metallic debris that frequently spray into the passenger behind the airbag. You discuss the results with your immediate supervisor who believes no action is required and notes "that this risk of death from a faulty airbag is still much less than the risk of death from being in an accident without any airbag." What should you do, if anything, about your supervisor's failure to act?

Scenario #2: You are a junior engineer in the Medical Products division of a large Fortune 100 company. Your division's most profitable product is a surgical hip implant that is primarily constructed out of titanium. Due to increased demand for titanium from aerospace manufacturers the price of the titanium alloy your company uses has increased by over 50% over the last two years. In order to reduce manufacturing costs your division has been ordered to shift to the use of titanium alloy Ti-6Al-4V vice the current alloy Ti-6Al-7Nb. Both alloys are commonly used in medical implants, though the current alloy is known to have better wear resistance and is slightly higher strength. One of the experienced technicians who manufacture the implants expresses his concern "…that implants made with the new alloy will not last as long and will be more likely to fracture in use..." You express the technician's concerns to your immediate supervisor, and he responds that "…both alloys are fine for this application…" and that he is under severe pressure to reduce manufacturing costs and to restore prior levels of profitability for this product. Are you comfortable shifting to the new alloy? If not what will you do?

Scenario #3: You have recently been hired as a junior engineer in software development for a social media company. One of the projects that you are working on includes improving the predictive text generator for the application. Your team has been tasked with integrating the new Open AI release GPT-2 which uses language models from the internet to predict text and can even generate "coherent paragraphs of text." While testing the implementation you notice that the predictive text can generate completely fictitious politically loaded narratives. You bring your concerns to your team leader, but they dismiss them to complete the project on time. Are you comfortable with your supervisor's decision? If not, what would be your next steps?

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Scenario #4: You are a junior civil engineer at a large construction firm that specializes in the design, construction or structural reinforcement of earthquake resistant buildings, bridges and other structures. The company is in California and has significant business due to recent complex code revisions to improve earthquake resistance of public structures, particularly hospitals. You are in an engineering group of five junior engineers who are responsible for the earthquake retrofit for a hospital located in Palo Alto, near San Francisco. Your engineering group's immediate supervisor is a Registered Professional Engineer (Civil Engineering) in the state of California. Your group has spent many weeks designing a major structural reinforcement for the hospital and presents the design to your supervisor for approval. After a brief review the supervisor says the design is acceptable and approves the design. The supervisor then forwards the design to the company's cost estimation branch so that the job can promptly commence once the hospital concurs with the cost of the retrofit. Are you comfortable with the rapidity that the retrofits were approved and provided to the cost estimation branch? If not, what should you do? Are there any related legal issues or concerns that need to be addressed?

Scenario #5: You are a junior engineer for a large automotive supplier. Your company was asked to develop software that would allow the automotive manufacturer to pass government emissions testing for various high efficiency engines. The software is written such that it provides different performance during emissions testing than it does during actual engine operation. During actual operation the software allows the engines to produce significantly more NO_X emissions than allowed by law. You later learn that the automotive manufacturer installs the software in vehicles sold to consumers. Your company's leadership is aware of this issue and informs the automotive manufacturer, via letter, that the software was not intended for this purpose. The automotive manufacturer ignores the notification and continues to sell vehicles with the suspect software. Your company's leadership decides that no additional action is required since it has informed the automotive manufacturer of the problem and is not liable for misuse of the software. Do you agree that no further action is required? Does your company have liability for the misuse of the software? What should you do?